

```
%DAY 1 (5 june 2026) Internship CoE-mathworks deeptech labs
%https://in.mathworks.com/help/matlab/getting-started-with-matlab.html?s_tid=CRUX_lftnav
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%reg no: 2023BEEE07AED030
%2023-27 batch %
%VII sem
%Alliance university
```

Desktop Basics

```
% 1st part
clc; clear
all; close
all; a=5;
b=6; c=[1
4 6]
```

```
c = 1×3
1     4     6
```

```
%2nd part
d=[1,2;2,2]
```

```
d = 2×2
1     2
2     2
```

```
a=sin(d)
```

```
a = 2×2
    0.8415    0.9093
    0.9093    0.9093
```

```
A=10;
```

```
t=[0,1,2,3,4,5,6,7,8,9,10]
```

```
t = 1×11
    0     1     2     3     4     5     6     7     8     9    10
```

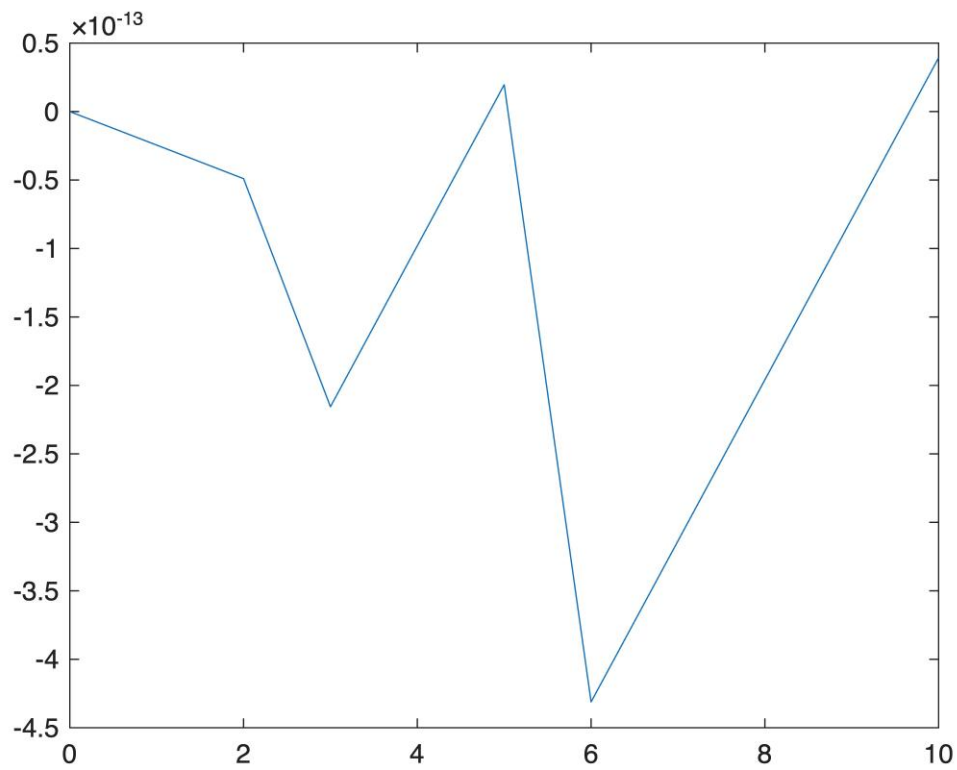
```
f=10
```

```
f = 10
```

```
ms=A*sin(2*pi*f*t)
```

```
ms = 1×11 10-
12 x
    0   -0.0245   -0.0490   -0.2156   -0.0980    0.0196   -0.4312   -0.3136 ...
```

```
plot(t,ms) %command for plotting
```



Matrices and Arrays

```
%array creation
```

```
a=[1 2 3 4]
```

```
a = 1×4
```

```
1      2      3      4
```

```
%Matrix creation 3×3
```

```
b=[1 3 5;3 4 6;1 2 5]
```

```
b = 3×3      1
      3      5
      3      4      6
1      2      5
```

```
c= [1 3 5      2 4 6
      7 8 10]
```

```
c = 3x3      1
      3      5      2
      4      6
           7      8      10
```

```
%4x4 matrix
```

```
a=[1 2 3 4; 2 3 4 5; 1 2 3 4;7 8 9 0]
```

```
a = 4x4
      1      2      3      4
      2      3      4      5
      1      2      3      4
      7      8      9      0
```

```
%only zeros matrix (5x5)
```

```
z1=zeros(5,5)
```

```
z1 = 5x5
      0      0      0      0      0
      0      0      0      0      0
      0      0      0      0      0
      0      0      0      0      0
      0      0      0      0      0
```

```
%only ones matrix (5x5)
```

```
z=ones(5,5)
```

```
z = 5x5
      1      1      1      1      1
      1      1      1      1      1
      1      1      1      1      1
      1      1      1      1      1
      1      1      1      1      1
```

Matrix and Array Operations

```
z+10 %addition
```

```
ans = 5x5
      11      11      11      11      11
      11      11      11      11      11
      11      11      11      11      11
      11      11      11      11      11
      11      11      11      11      11
```

```
z-10 %subtraction
```

```
ans = 5x5
      -9      -9      -9      -9      -9
```

```

-9    -9    -9    -9    -9
-9    -9    -9    -9    -9
-9    -9    -9    -9    -9
-9    -9    -9    -9    -9

```

```
z*10 %multiplication
```

```

ans = 5×5
-10    10    10    10    10
    10    10    10    10    10
    10    10    10    10    10
    10    10    10    10    10
    10    10    10    10    10

```

```
z/10 %division
```

```

ans = 5×5
    0.1000    0.1000    0.1000    0.1000    0.1000
    0.1000    0.1000    0.1000    0.1000    0.1000
    0.1000    0.1000    0.1000    0.1000    0.1000
    0.1000    0.1000    0.1000    0.1000    0.1000
    0.1000    0.1000    0.1000    0.1000    0.1000

```

```
%inverse function
```

```
p = a*inv(a)
```

```
Warning: Matrix is singular to working precision.
```

```

p = 4×4    Inf
Inf    I
    Inf    Inf    I
    Inf    Inf    I
    NaN    NaN    N

```

```
p = a.*a
```

```

p = 4×4
    1     4     9    16
    4     9    16    25
    1     4     9    16
   49    64    81     0

```

```
a.^3 %power
```

```

ans = 4×4    1     8
   27    64
    8    27    64   125
    1     8    27    64
   343   512   729     0

```

```
a.^5
```

```

ans = 4×4
    1     32    243   1024

```

32	243	1024	3125
1	32	243	1024
16807	32768	59049	0

```
%Concatenation
```

```
A = [a,a]
```

```
A = 4x8
```

1	2	3	4	1	2	3	4
2	3	4	5	2	3	4	5
	1	2	3	4	1	2	3
	7	8	9	0	7	8	9
							4
							0

```
A = [a; a]
```

```
A = 8x4
```

1	2	3	4
2	3	4	5
	1	2	3
	7	8	9
			4
			0
1	2	3	4
2	3	4	5
	1	2	3
	7	8	9
			4
			0

```
%complex numbers sqrt(-1)
%square root of -1
```

```
ans = 0.0000 +
1.0000i
```

```
c = [3+4i, 4+3j; -i, 10j] %matrix
```

```
c = 2x2 complex
3.0000 + 4.0000i 4.0000 + 3.0000i
0.0000 - 1.0000i 0.0000 +10.0000i
```

Text and Characters

```
%Text in String Arrays
t= "hello, world";
```

```
whos t %identity of t
```

Name	Size	Bytes	Class	Attributes
t	1x1	176	string	

```
f = 71; c =
(f-32)/1.8;
tempText = "Temperature is " + c + "C" %concatenation
```

```
tempText =
"Temperature is 21.6667C"
```

```
A = ["a", "bb", "ccc"; "dddd", "eeeeee", "ffffff"]
```

```
A = 2×3 string
```

```
"a"      "bb"      "ccc"  
"dddd"   "eeeeee"  "ffffff"
```

```
strlength(A) %command for length of a string
```

```
ans = 2×3
```

```
1    2    3  
4    6    7
```

Data in Character Arrays

```
h='huzefa p'; whos h  
%identity of h
```

Name	Size	Bytes	Class	Attributes
h	1×8	16	char	

```
h(5) %single character.
```

```
ans = 'f'
```

```
h2=[h 'hello'] %Concatenate character arrays with square brackets
```

```
h2 = 'huzefa  
phello'
```

Calling Functions

```
max(a) %max of matrix (a)
```

```
ans = 1×4
```

```
7    8    9    5
```

```
a=[1 3 5];  
max(a)
```

```
ans = 5
```

```
% maximum of (a)
```

```
b=[3 5 6];  
union(a,b) %union of a and b
```

```
ans = 1×4
```

```
1    3    5    6
```

```
max(b)
```

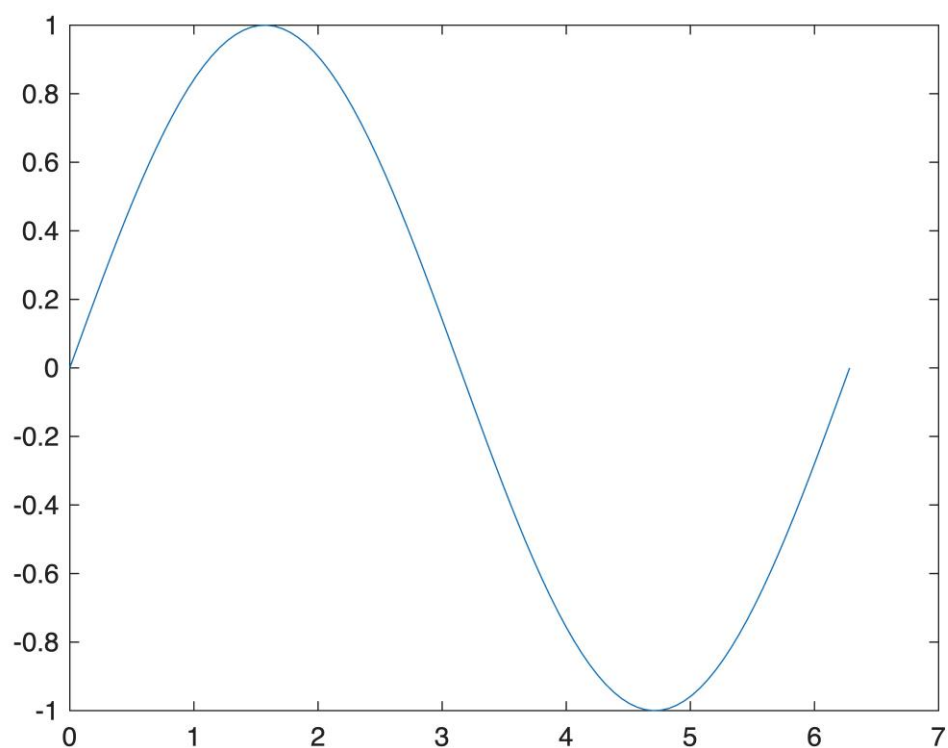
```
ans = 6
```

```
disp("hello world") %display or print
```

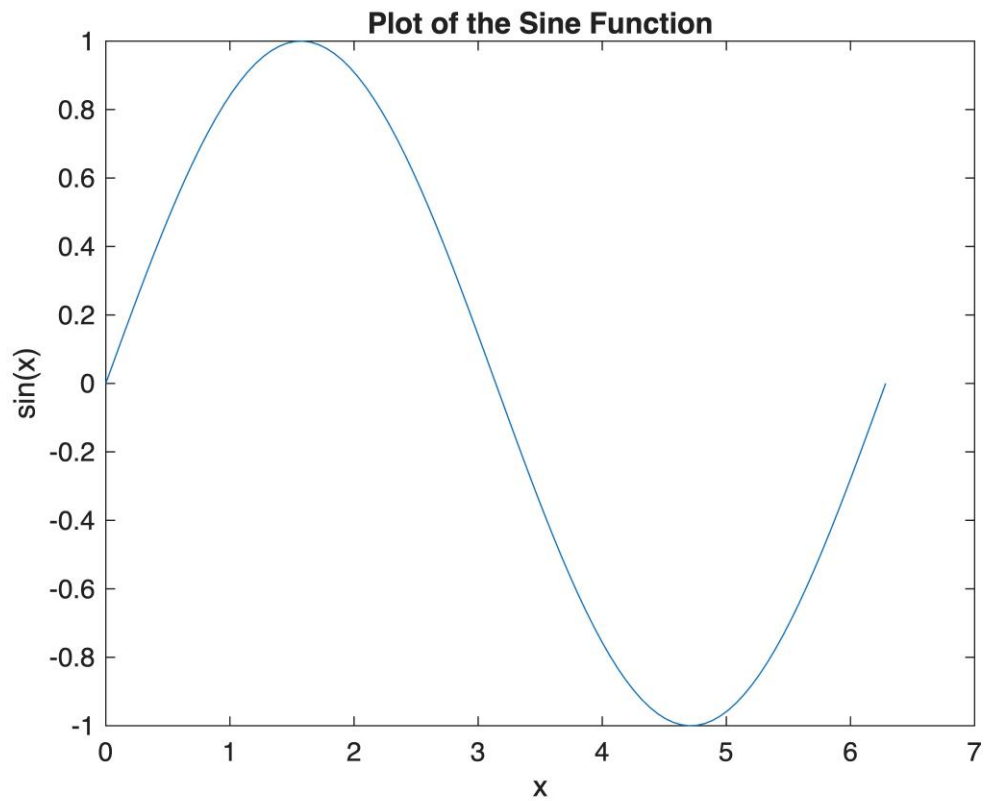
```
hello world
```

2-D and 3-D Plots

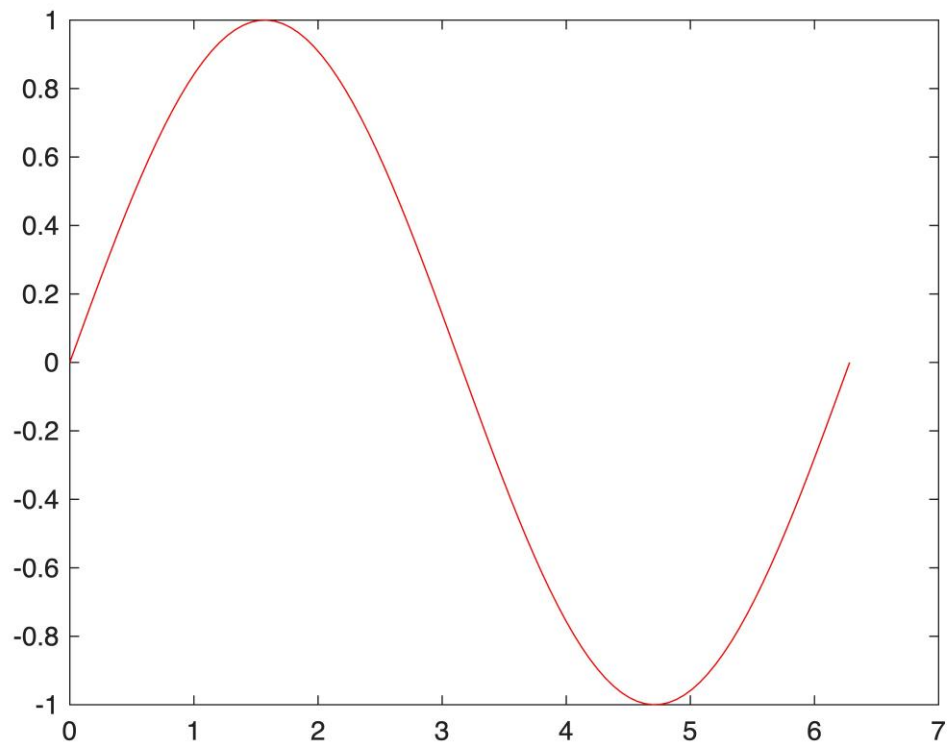
```
%Line Plots  
x=linspace(0,2*pi); y=sin(x);  
plot(x,y) %command of plotting
```



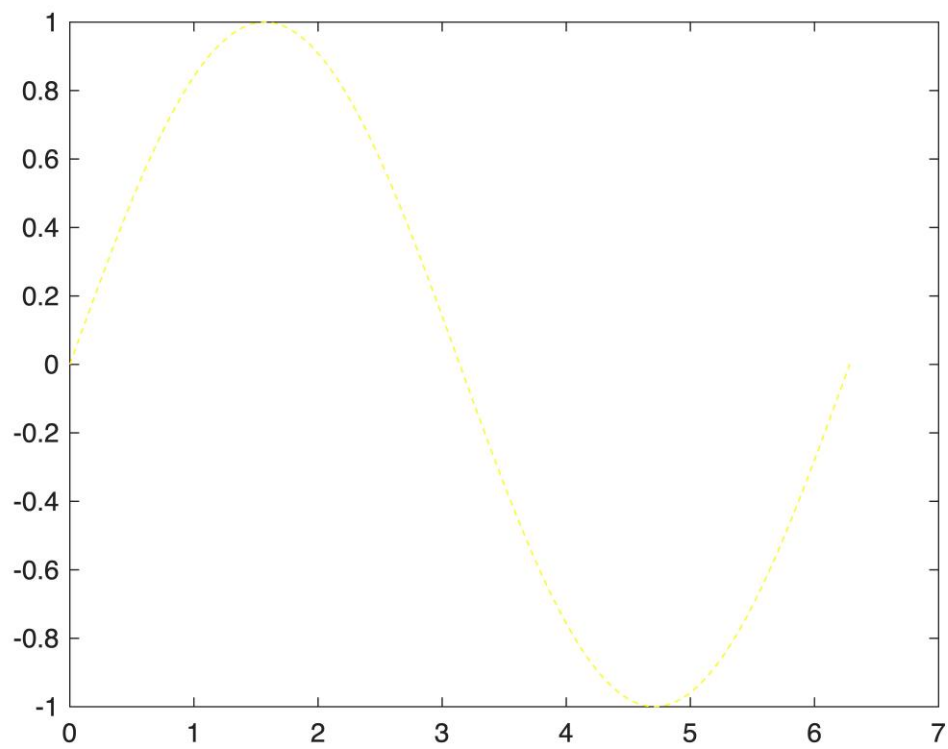
```
%giving title  
xlabel("x")  
ylabel("sin(x)")  
title("Plot of the Sine Function")
```



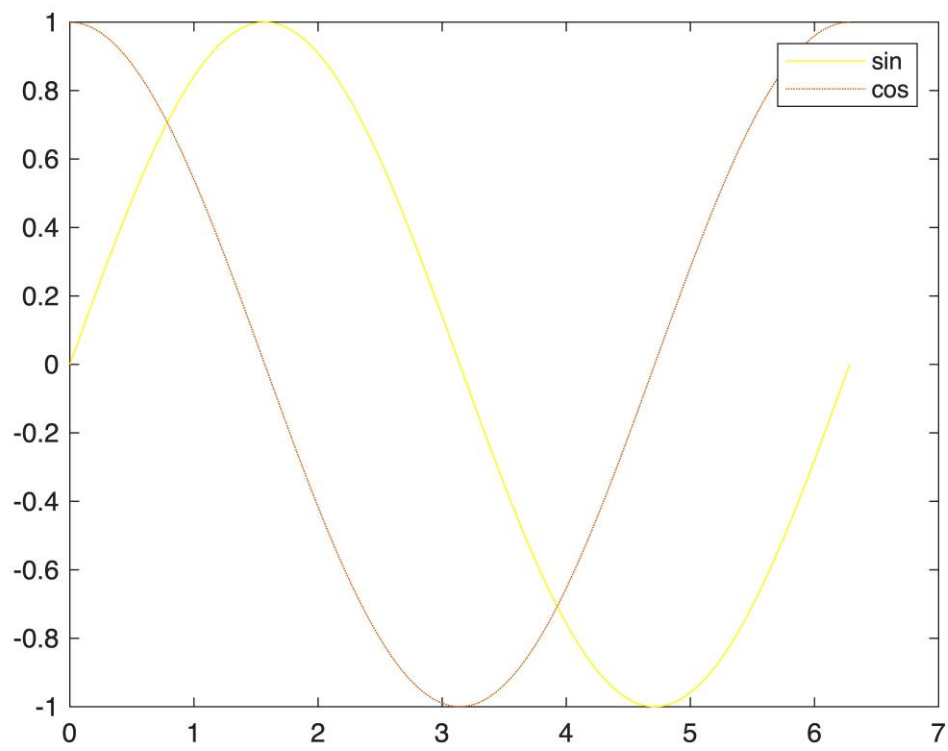
```
plot(x,y,"r") %command for colour (r=red)
```



```
plot(x,y,"y--") %use -- for dotted line %command for colour (y=yellow)
```

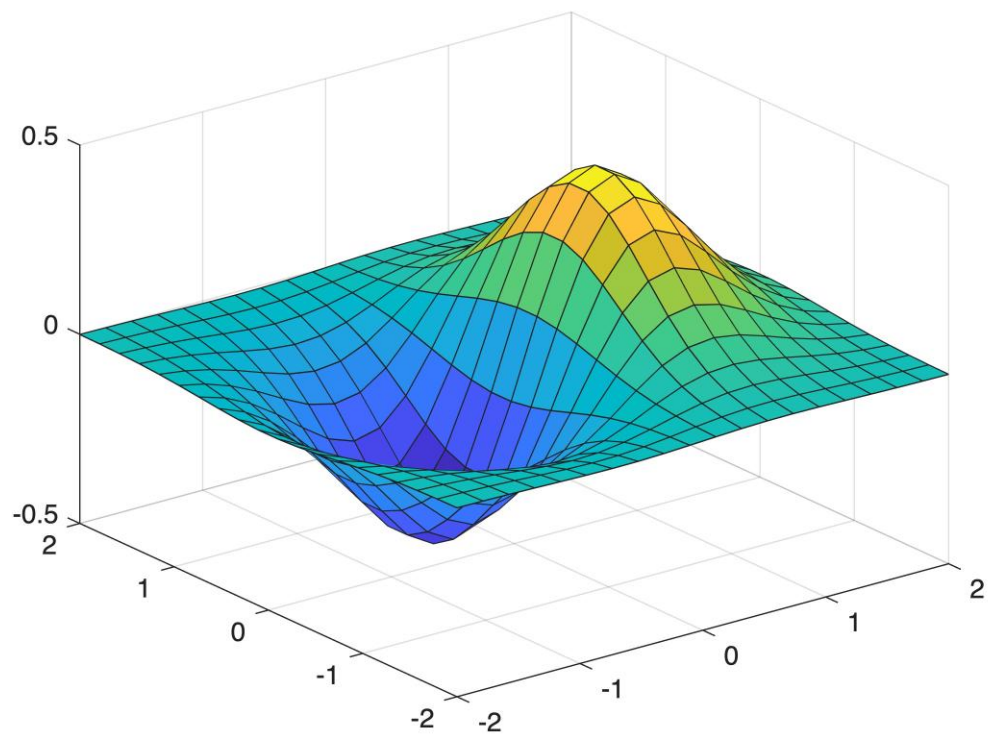



```
x = linspace(0,2*pi); y = sin(x);  
plot(x,y,"y") %"y" for colour (yellow)  
  
hold on  
  
y2 = cos(x); plot(x,y2,":") %use : for  
dotted lines legend("sin","cos") hold  
off
```

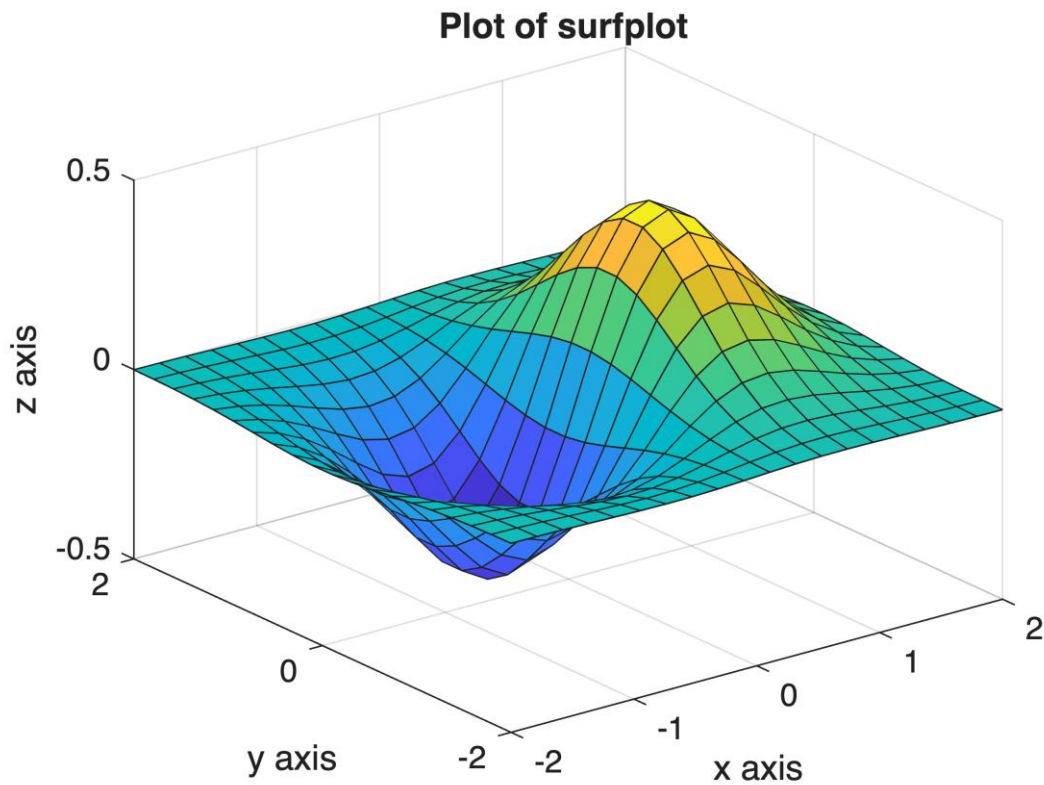


3-D Plots

```
x=linspace(-2,2,20);  
y=x'; z=x.*exp(-x.^2-  
y.^2); surf(x,y,z)
```



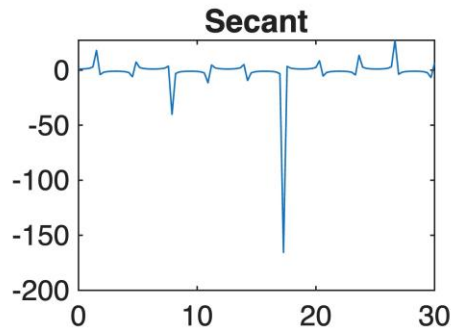
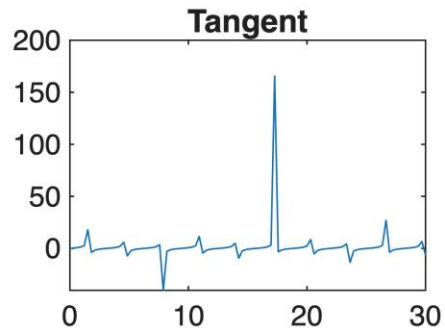
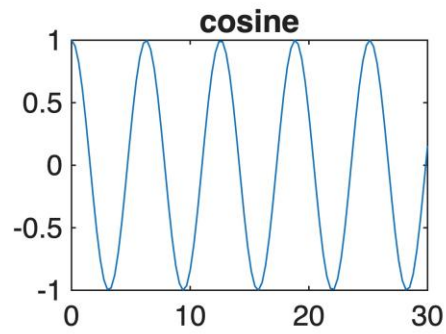
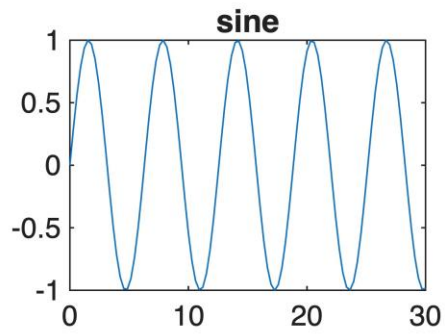
```
%giving title and labels  
xlabel("x axis") ylabel("y  
axis") zlabel("z axis")  
title("Plot of surfplot ")
```



Multiple Plots

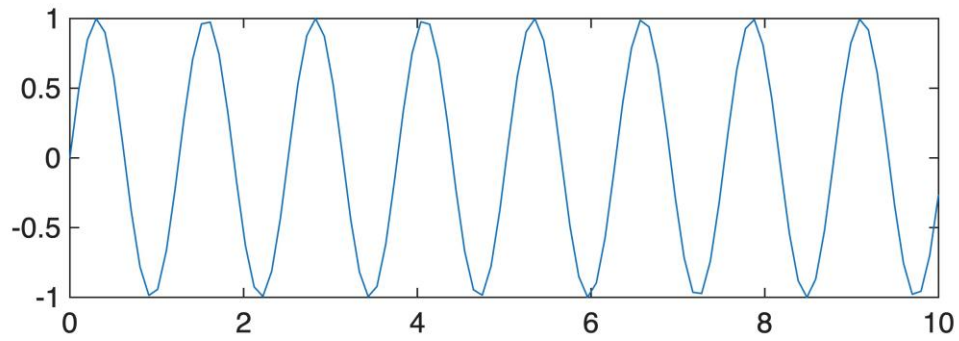
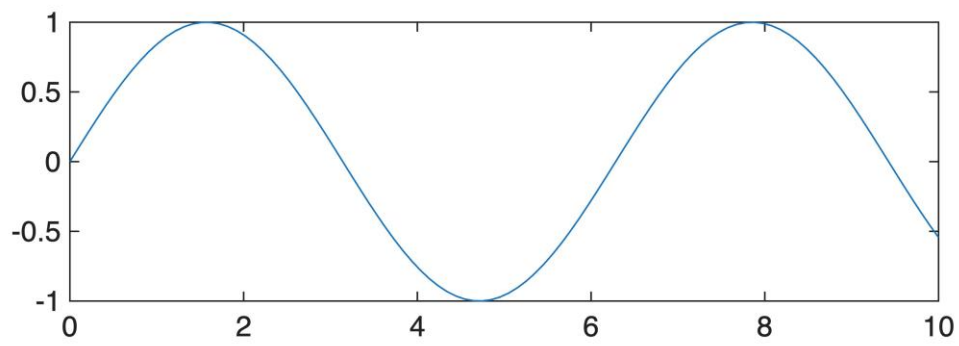
```
t = tiledlayout(2,2);  
title(t,"Trigonometric functions")  
x=linspace(0,30);  
  
nexttile  
plot(x,sin(x))  
title("sine")  
  
nexttile  
plot(x,cos(x))  
title("cosine")  
  
nexttile  
plot(x,tan(x))  
title("Tangent")  
  
nexttile  
plot(x,sec(x))  
title("Secant")
```

Trigonometric functions



Subplotting

```
subplot(2,1,1);  
x=linspace(0,10);  
y1=sin(x); plot(x,y1)  
  
subplot(2,1,2);  
y2=sin(5*x);  
plot(x,y2)
```



```
%stem function
```

```
x=1:10
```

```
x = 1×10
```

```
1    2    3    4    5    6    7    8    9   10
```

```
y=1:10
```

```
y = 1×10
```

```
1    2    3    4    5    6    7    8    9   10
```

```
stem(x,y)
```

